# 4.6 Multivariable Stochastic Calculus

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# 4.6.1 Multiple Brownian Motion

**Definition 4.6.1.** A d-dimensional Brownian motion is a process $W(t) = (W_1(t), \ldots, W_n(t))$ with the following properties.

(1) Each $W_i(t)$ is a one-dimensional Brownian motion.

(2) If $i \neq j$, then the processes $W_i(t)$ and $W_j(t)$ are independent.

Associated with a d-dimensional Brownian motion, we have a filtration $F(t)$, $t \geq 0$, such that the following holds.

(3) (**Information accumulates**) For $0 \leq s < t$, every set in $F(s)$ is also in $F(t)$.

(4) (**Adaptivity**) For each $t \geq 0$, the random vector $W(t)$ is $F(t)$-measurable.

(5) (**Independence of future increments**) For $0 \leq t < u$, the vector of increments $W(u)-W(t)$ is independent of $F(t)$.

- Each component of a d-dimension Brownian motion is a one-dimensional Brownian motion.

1. quadratic variation formula $[W_i, W_i](t) = t$, and writing informally as

$$dW_i(t)\,dW_i(t) = dt.$$

2. If $i \neq j$, the independence of the $W_i(t)$ and $W_j(t)$ implies $\left[W_i, W_j\right](t) = 0$ and writing informally as

$$dW_i(t)\,dW_j(t) = 0, i \neq j.$$

- Let $\Pi = \{t_0, \dots, t_n\}$ be a partition of [0,T]. For $i \neq j$, define the sampled cross variation of $W_i$ and $W_j$ on [0,T] to be

$$C_\Pi = \sum_{k=0}^{n-1} \left[ W_i(t_{k+1}) - W_i(t_k) \right] \left[ W_j(t_{k+1}) - W_j(t_k) \right].$$

- The increment are independent of one another and have mean zero $\left( \mathrm{E} C_\Pi = 0 \right)$.

$$\begin{aligned} Var\left(C_\Pi\right) = EC_\Pi^2 &= \sum_{k=0}^{n-1} \left[ W_i(t_{k+1}) - W_i(t_k) \right]^2 \left[ W_j(t_{k+1}) - W_j(t_k) \right]^2 \\ &= \sum_{k=0}^{n-1} \left( t_{k+1} - t_k \right)^2 \\ &\le \|\Pi\| \cdot \sum_{k=0}^{n-1} \left( t_{k+1} - t_k \right) = \|\Pi\| \cdot T \end{aligned}$$

- As $\|\Pi\| \to 0$, we have $Var\left(C_\Pi\right) \to 0$, so $C_\Pi$ converges to the constant $EC_\Pi = 0$.

# 4.6.2 Ito-Doeblin Formula for Multiple Processes

- Let X(t) and Y(t) be Ito processes, then

$$X(t) = X(0) + \int_0^t \Theta_1(u)\,du + \int_0^t \sigma_{11}(u)\,dW_1(u) + \int_0^t \sigma_{12}(u)\,dW_2(u),$$

$$Y(t) = Y(0) + \int_0^t \Theta_2(u)\,du + \int_0^t \sigma_{21}(u)\,dW_1(u) + \int_0^t \sigma_{22}(u)\,dW_2(u).$$

- Assumed the integrands $\Theta_i(u)$ and $\sigma_{ij}(u)$ are adapted processes. In differential notation,

$$dX(t) = \Theta_1(t)\,dt + \sigma_{11}(t)\,dW_1(t) + \sigma_{12}(t)\,dW_2(t), \qquad (4.6.1)$$

$$dY(t) = \Theta_2(t)\,dt + \sigma_{21}(t)\,dW_1(t) + \sigma_{22}(t)\,dW_2(t). \qquad (4.6.2)$$

- The Ito integral $\int_0^t \sigma_{11}(u)dW_1(u)$ accumulates quadratic variation at rate $\sigma_{11}^2(t)$ per unit time.

- And the Ito integral $\int_0^t \sigma_{12}(u)dW_2(u)$ accumulates quadratic variation at rate $\sigma_{12}^2(t)$ per unit time.

- Because both of these integrals appear in $X(t)$, the process X(t) accumulates quadratic variation at rate $\sigma_{11}^2(t)+\sigma_{12}^2(t)$ per unit time:

$$[X,X](t)=\int_0^t\left(\sigma_{11}^2(u)+\sigma_{12}^2(u)\right)du.$$

- In differential form:

$$dX(t)\,dX(t)=\left(\sigma_{11}^2(t)+\sigma_{12}^2(t)\right)dt. \qquad (4.6.3)$$

Using the multiplication rules

$$dt\,dt=0,\ dt\,dW_i(t)=0,\ dW_i(t)\,dW_i(t)=dt,$$

$$dW_i(t)\,dW_j(t)=0 \ \text{for } i\neq j$$

- In a similar way,

$$dY(t)\,dY(t)=\left(\sigma_{21}^2(t)+\sigma_{22}^2(t)\right)dt. \qquad (4.6.4)$$

$$dX(t)\,dY(t)=\left(\sigma_{11}(t)\sigma_{21}(t)+\sigma_{12}(t)\sigma_{22}(t)\right)dt. \qquad (4.6.5)$$

- Above equation says that, for every $T\geq 0$,

$$[X,Y](t)=\int_0^T\left(\sigma_{11}(t)\sigma_{21}(t)+\sigma_{12}(t)\sigma_{22}(t)\right)dt. \qquad (4.6.6)$$

- The term is defined as follow. Let $\Pi = \{t_0, \quad , t_n\}$ be a partition of [0,T] and set up the sampled cross variation

$$\sum_{k=0}^{n-1} \left[ X(t_{k+1}) - X(t_k) \right] \left[ Y(t_{k+1}) - Y(t_k) \right].$$

- When n go to infinity as the length of the longest subinterval $\|\Pi\| = \max_{0 \le k \le n-1} (t_{k+1} - t_k)$ goes to zero. This limit of the sum is $[X, Y](t)$.

- The proof is similar to the proof Lemma 4.4.4, with the additional feature $[W_1, W_2](t) = 0.$

- **Theorem 4.6.2 (Two-dimensional Ito-Doeblin formula).**
  Let *f(t, x, y)* be a function whose partial derivatives $f_t, f_x, f_y, f_{xx}, f_{xy}, f_{yx}$ *and* $f_{yy}$ are defined and are continuous.

  Let $X(t)$ and $Y(t)$ be Ito processes as discussed above.
  The two-dimensional Ito-Doblen formula in differential form is

$$
\begin{aligned}
df\left(t, X(t), Y(t)\right) = {} & f_t\left(t, X(t), Y(t)\right)dt + f_x\left(t, X(t), Y(t)\right)dX(t) \\
& + f_y\left(t, X(t), Y(t)\right)dY(t) + \\
& \frac{1}{2} f_{xx}\left(t, X(t), Y(t)\right)dX(t)dX(t) \\
& + f_{xy}\left(t, X(t), Y(t)\right)dX(t)dY(t) + \\
& \frac{1}{2} f_{yy}\left(t, X(t), Y(t)\right)dY(t)dY(t)
\end{aligned}
$$

- The proof of the theorem 4.6.2 is similar to that of theorem 4.4.6.

- Rewrite above equation leaving out t. The compact notation :

$$df\left(t, X, Y\right) = f_t dt + f_x dX + f_y dY + \frac{1}{2} f_{xx} dXdX + f_{xy} dXdY + \frac{1}{2} f_{yy} dYdY \qquad (4.6.9)$$

- The right-hand side of the above the equation is the Taylor series expansion of *f* out second order. The full the Taylor series expansion:

$$df\left(t,X,Y\right)=f_t dt+f_x dX+f_y dY+\frac{1}{2}f_{tt}dtdt+\frac{1}{2}f_{xx}dXdX+$$

$$\frac{1}{2}f_{yy}dYdY+\frac{1}{2}f_{xy}dXdY+\frac{1}{2}f_{yx}dYdX+\frac{1}{2}f_{xt}dXdt$$

$$+\frac{1}{2}f_{tx}dtdX+\frac{1}{2}f_{ty}dtdY+\frac{1}{2}f_{ty}dYdt$$

- But *dtdt, dtdX*, and *dtdY* are zero. The function *f* whose second partial derivatives exist and are continuous($f_{xy}=f_{yx}$), and so we have combined these terms into the single term $\left(f_{xy}dXdY\right)$.

- Making (4.6.1)-(4.6.5) substitution and then integration(4.6.9), we obtain the Ito-Doeblin formula in integral form:

$$
\begin{aligned}
& f\left(t, x(t), Y(t)\right) - f\left(0, x(0), Y(0)\right) \\
&= \int_0^t \left[ \sigma_{11}(u) f_x\left(u, X(u), Y(u)\right) + \sigma_{21}(u) f_y\left(u, X(u), Y(u)\right) \right] dW_1(u) + \\
&\quad \int_0^t \left[ \sigma_{12}(u) f_x\left(u, X(u), Y(u)\right) + \sigma_{22}(u) f_y\left(u, X(u), Y(u)\right) \right] dW_2(u) + \\
&\quad \int_0^t \Big[ f_t\left(u, X(u), Y(u)\right) + \Theta_1(u) f_x\left(u, X(u), Y(u)\right) + \\
&\quad \Theta_2(u) f_y\left(u, X(u), Y(u)\right) + \\
&\quad \frac{1}{2}\left(\sigma_{11}^2(u) + \sigma_{12}^2(u)\right) f_{xx}\left(u, X(u), Y(u)\right) + \\
&\quad \left(\sigma_{11}(u)\sigma_{21}(u) + \sigma_{12}(u)\sigma_{22}(u)\right) f_{xy}\left(u, X(u), Y(u)\right) + \\
&\quad \frac{1}{2}\left(\sigma_{21}^2(u) + \sigma_{22}^2(u)\right) f_{yy}\left(u, X(u), Y(u)\right) \Big] du.
\end{aligned}
$$

- The right-hand side of this equation has one ordinary (Lebesgue) integral with respect to *du* and two Ito integrals with respect to $dW_1(u)$ and $dW_2(u)$.

- **Corollary 4.6.3 (Ito product rule)**

  Let $X(t)$ and $Y(t)$ be Ito processes. Then

$$d\left(X(t)Y(t)\right) = X(t)\,dY(t) + Y(t)\,dX(t) + dX(t)\,dY(t).$$

Proof:

Take $f(t, x, y)=xy$, so that $f_t=0$, $f_x=y$, $f_y=x$, $f_{xx}=0$, $f_{xy}=1$, and $f_{yy}=0$.

Then by (4.6.9), $df=d(xy)=ydx+xdy+dxdy$

# 4.6.3 Recognizing a Brownian Motion

- **Theorem 4.6.4(Levy, one dimension)**

Let $M(t),\, t \geq 0,$ be a martingale relative to a filtration $F(t), t \geq 0.$

Assume that:

- 1.M(0)=0
- 2.M(t) has continuous paths
- 3.[M,M](t)=t for all $t \geq 0.$

Then M(t) is a Brownian motion.

- Idea of the proof:

A Brownian motion is martingale whose increments are normally distributed. Levy's theorem do not assume the normality.

- The method used to establish normality is to first check that in the derivation of the Ito-Doeblin formula for Brownian motion. Theorem 4.4.1 assumed quadratic variation $[M,M](t)=t$ in this theorem is a continuous process.

- Doeblin formula may be applied to $M$ with the result that, for any function $f(t,x)$ whose derivatives exist and are continuous.

$$df\left(t,M\left(t\right)\right)=f_t\left(t,M\left(t\right)\right)dt+f_x\left(t,M\left(t\right)\right)dM\left(t\right)+\frac{1}{2}f_{xx}\left(t,M\left(t\right)\right)dt$$

- In integrated form,

$$f\left(t,M\left(t\right)\right)=f\left(0,M\left(0\right)\right)+\int_0^t\left[f_t\left(s,M\left(s\right)\right)+\frac{1}{2}f_{xx}\left(s,M\left(s\right)\right)\right]ds$$

$$+\int_0^t f_x\left(s,M\left(s\right)\right)dM\left(s\right)$$

- Because *M*(*t*) is martingale, the stochastic integral $\int_0^t f_x\left(s,M\left(s\right)\right)dM\left(s\right)$ is also.

At t=0, this stochastic integral takes the value zero, so its expectation is zero.

$$Ef\left(t,M\left(t\right)\right)=f\left(0,M\left(0\right)\right)+E\int_0^t\left[f_t\left(s,M\left(s\right)\right)+\frac{1}{2}f_{xx}\left(s,M\left(s\right)\right)\right]ds$$

$$Ef\left(t,M\left(t\right)\right)=f\left(0,M\left(0\right)\right)+E\int_0^t\left[f_t\left(s,M\left(s\right)\right)+\frac{1}{2}f_{xx}\left(s,M\left(s\right)\right)\right]ds$$

- Define $f\left(t,x\right)=\exp\left\{ux-\frac{1}{2}u^2t\right\}$, a fixed number $u$
- Then $f_t\left(t,x\right)=-\frac{1}{2}u^2f\left(t,x\right), f_x\left(t,x\right)=uf\left(t,x\right),$

  and $f_{xx}\left(t,x\right)=u^2f\left(t,x\right).$
- In particular, $f_t\left(t,x\right)+\frac{1}{2}f_{xx}\left(t,x\right)=0.$

$$\boldsymbol{E}exp\left\{uM(t)-\frac{1}{2}u^2t\right\}=f\big(0,M(0)\big)=\exp(0)=1 \Rightarrow \boldsymbol{E}e^{uM(t)}=e^{\frac{1}{2}u^2t}$$

- This is the moment-generating function for the normal distribution with mean zero and variance t. $M(t)$ is normality distribution.

- **Theorem 4.6.5(Levy, two dimension)**

Let. $M_1(t)$ and $M_2(t)$, $t \geq 0$, be a martingale relative to a filtration $F(t), t \geq 0$.
Assume that for i = 1,2, we have :

- 1. $M_i(0)=0$
- 2. $M_i(t)$ has continuous paths
- 3. $[M_i,M_i](t)=t$ for all $t \geq 0$.

In addition, $[M_1,M_2](t)=0$ for all $t \geq 0$

Then $M_1(t)$ and $M_2(t)$ are independence Brownian motions.

- Idea of the proof:

The Theorem 4.6.4 implies that $M_1$ and $M_2$ are Brownian motions.

- Claim: independence

Let $f(t,x,y)$ be a derivative and continuous function. The two-dimension Ito-Doblin formula implies that

$$
\begin{aligned}
df\left(t,M_1,M_2\right) &= f_t dt + f_x dM_1 + f_y dM_2 + \\
&\quad \frac{1}{2} f_{xx} dM_1 dM_1 + f_{xy} dM_1 dM_2 + \frac{1}{2} f_{yy} dM_2 dM_2 \\
&= f_t dt + f_x dM_1 + f_y dM_2 + \frac{1}{2} f_{xx} dt + \frac{1}{2} f_{yy} dt
\end{aligned}
$$

- where assumption, $\left[M_1, M_1\right](t) = t \quad \Rightarrow dM_1 dM_1 = dt$

$$\left[M_2, M_2\right](t) = t \quad \Rightarrow dM_2 dM_2 = dt$$

$$\left[M_1, M_2\right](t) = 0 \quad \Rightarrow dM_1 dM_2 = 0$$

- Integration

$$\begin{aligned} f\left(t, M_1(t), M_2(t)\right) = f\left(0, M_1(0), M_2(0)\right) + \int_0^t \Big[ f_t\left(s, M_1(s), M_2(s)\right) + \\ \frac{1}{2} f_{xx}\left(s, M_1(s), M_2(s)\right) + \frac{1}{2} f_{yy}\left(s, M_1(s), M_2(s)\right) \Big] ds \\ + \int_0^t f_x\left(s, M_1(s), M_2(s)\right) dM_1(s) \\ + \int_0^t f_y\left(s, M_1(s), M_2(s)\right) dM_2(s) \end{aligned}$$

- The last two terms on the right-hand side are martingale, starting at zero at time zero, and expectation zero.

$$Ef\left(t, M_1(t), M_2(t)\right) = f\left(0, M_1(0), M_2(0)\right) + E\int_0^t \Big[ f_t\left(s, M_1(s), M_2(s)\right) + \frac{1}{2} f_{xx}\left(s, M_1(s), M_2(s)\right) + \frac{1}{2} f_{yy}\left(s, M_1(s), M_2(s)\right) \Big] ds$$

- Define

$$f(t,x,y) = \exp\left\{ u_1 x + u_2 y - \frac{1}{2}\left(u_1^2 + u_2^2\right)t \right\}, \text{a fixed number } u_1 \text{ and } u_2$$

- Then, $f_t(t,x,y) = -\frac{1}{2}\left(u_1^2 + u_2^2\right) f(t,x,y),\ f_x(t,x,y) = u_1 f(t,x,y),$

$$f_y(t,x,y) = u_2 f(t,x,y),\ f_{xx}(t,x,y) = u_1^2 f(t,x,y), \text{ and}$$

$$f_{yy}(t,x,y) = u_2^2 f(t,x,y),$$

- In particular,

$$f_t\left(s,M_1(s),M_2(s)\right)+\frac{1}{2}f_{xx}\left(s,M_1(s),M_2(s)\right)+\frac{1}{2}f_{yy}\left(s,M_1(s),M_2(s)\right)=0.$$

$$E\exp\left\{u_1M_1(t)+u_2M_2(t)-\frac{1}{2}\left(u_1^2+u_2^2\right)t\right\}=1.$$

$$\Rightarrow Ee^{u_1M_1(t)+u_2M_2(t)}=e^{\frac{1}{2}u_1^2t}\cdot e^{\frac{1}{2}u_2^2t}.$$

- The joint moment-generating function factors into the product of moment-generating functions, $M_1$ and $M_2$ must be independence.

- **Example 4.6.4(Correlated stock prices)**

  Suppose $\dfrac{dS_1(t)}{S_1(t)} = \alpha_1 dt + \sigma_1 dW_1(t)$

$$\frac{dS_2(t)}{S_2(t)} = \alpha_2 dt + \sigma_2\left[\rho dW_1(t) + \sqrt{1-\rho^2}\, dW_2(t)\right]$$

Where $W_1(t)$ and $W_2(t)$ are independent Brownian motion and $\sigma_1 > 0, \sigma_2 > 0$ and $-1 \le \rho \le 1$ are constant.

To analysis the stock price $S_2$ process, we define $W_3(t) = \rho W_1(t) + \sqrt{1-\rho^2}\, W_2(t)$.
$W_3(t)$ is a continuous martingale with $W_3(0) = 0$.

$$
\begin{aligned}
dW_3 dW_3 &= \rho^2 dW_1(t)\,dW_1(t) + 2\rho\sqrt{1-\rho^2}\,dW_1(t)\,dW_2(t) \\
&\quad + \left(1-\rho^2\right) dW_2(t)\,dW_2(t) \\
&= \rho^2 dt + \left(1-\rho^2\right) dt = dt \\
\Rightarrow \left[W_3, W_3\right] &= t
\end{aligned}
$$

- By one-dimension Levy Theorem, Theorem4.6.4, $W_3(t)$ is a Brownian motion.

- The differential of $S_2(t)$ as $\frac{dS_2(t)}{S_2(t)} = \alpha_2 dt + \sigma_2 dW_3$, is a geometric Brownian motion with mean rate of return $\alpha_2$ and volatility $\sigma_2$.

- The Brownian motion $W_1(t)$ and $W_3(t)$ are correlated. According to Ito' product rule,

$$d\left(W_1(t)W_3(t)\right) = W_1(t)\,dW_3(t) + W_3(t)\,dW_1(t) + dW_1(t)\,dW_3(t)$$
$$= W_1(t)\,dW_3(t) + W_3(t)\,dW_1(t) + \rho dt$$

- Integrating, $W_1(t)W_3(t) = \int_0^t W_1(t)\,dW_3(t) + \int_0^t W_3(t)\,dW_1(t) + \rho t.$

- The Ito integrals have expectation zero, so the covariance of $W_1(t)$ and $W_3(t)$ is

$$E\left[W_1(t)W_3(t)\right] = \rho t.$$

- Because $W_1(t)$ and $W_3(t)$ have standard deviation $\sqrt{t}$ , the number $\rho$ is the correlation between $W_1(t)$ and $W_3(t)$.